

Workshop 1 Oral Q&A — Precise Solutions

ELEN90098 Reinforcement Learning for Engineering

How to use these notes

The oral questions are expected to ask *how we answered the workshop questions*. For each question, the focus is therefore

what was asked $\rightarrow$ method $\rightarrow$ implementation $\rightarrow$ result $\rightarrow$ interpretation

Important update for Q3: $\epsilon = 0.01$ is classified as **inconclusive** from our finite-horizon numerical evidence.

Common experimental setup

We use a stationary 10-armed Gaussian bandit:

$$k = 10, \quad q^*(a) \sim \mathcal{N}(0, 1),$$

and

$$R_t \mid A_t = a \sim \mathcal{N}(q^*(a), 1).$$

Using seed 5 gives a fixed testbed for all algorithms. The optimal arm is

$$a^* = 2$$

in zero-based Python indexing, i.e. the third arm, with

$$q^*(a^*) = 2.4308.$$

Simulation settings:

Tasks	T	M
Q3	10^5	500
Q4–Q7	10^4	300

The true values $q^*(a)$ are known only to us as experimenters. They are used to evaluate the algorithms, but are never supplied to the learning policies.

Q1 — Define sublinear regret and explain its significance

The cumulative pseudo-regret is

$$\text{Regret}_T = \mathbb{E} \left[\sum_{t=0}^{T-1} (q^*(a^*) - q^*(A_t)) \right].$$

Sublinear regret means

$$\boxed{\text{Regret}_T = o(T)}$$

or equivalently

$$\boxed{\lim_{T \rightarrow \infty} \frac{\text{Regret}_T}{T} = 0.}$$

This means cumulative regret may still increase, but it grows slower than linearly.

Since

$$\frac{\text{Regret}_T}{T} = q^*(a^*) - \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[q^*(A_t)],$$

sublinear regret implies

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[q^*(A_t)] \rightarrow q^*(a^*).$$

Thus the long-run average expected reward approaches the optimal arm's expected reward.

Define the gap

$$\Delta_a = q^*(a^*) - q^*(a).$$

Then

$$\text{Regret}_T = \sum_{a \neq a^*} \Delta_a \mathbb{E}[N_T(a)].$$

Hence, for every strictly suboptimal arm,

$$\frac{\mathbb{E}[N_T(a)]}{T} \rightarrow 0.$$

Oral-ready answer. Sublinear regret means $\text{Regret}_T = o(T)$, equivalently $\text{Regret}_T / T \rightarrow 0$. So the average regret per decision vanishes, which means the long-run average expected reward approaches that of the optimal arm.

Q2 — Propose a numerical way to examine sublinear regret

Sublinearity is an asymptotic property, so a finite simulation cannot prove it. We therefore evaluate regret over increasing horizons and use three complementary diagnostics.

For M independent reruns, estimate cumulative pseudo-regret by

$$\widehat{R}_T = \frac{1}{M} \sum_{r=1}^M \sum_{t=0}^{T-1} \left[q^*(a^*) - q^*(A_t^{(r)}) \right].$$

We evaluate the policies over increasing horizons up to

$$T = 10^5.$$

1. Normalised regret

$$\widehat{\rho}_T = \frac{\widehat{R}_T}{T}.$$

This is the primary diagnostic because it directly corresponds to

$$\frac{\text{Regret}_T}{T} \rightarrow 0.$$

A sustained decrease toward zero is consistent with sublinear regret. A positive plateau is evidence of linear regret.

2. Incremental regret rate

Between consecutive horizons,

$$\widehat{s}_i = \frac{\widehat{R}_{T_{i+1}} - \widehat{R}_{T_i}}{T_{i+1} - T_i}.$$

This estimates the additional regret accumulated per additional time step over the interval.

If

$$R_T \approx cT,$$

then

$$\widehat{s}_i \approx c > 0.$$

A decreasing late-horizon incremental rate is therefore consistent with sublinear growth.

3. Effective exponent

Assume locally

$$\widehat{R}_T \approx CT^\alpha.$$

Taking logarithms,

$$\log \widehat{R}_T \approx \log C + \alpha \log T.$$

Hence

$$\widehat{\alpha}_i = \frac{\log \widehat{R}_{T_{i+1}} - \log \widehat{R}_{T_i}}{\log T_{i+1} - \log T_i}.$$

Interpretation:

$$\hat{\alpha}_i < 1$$

is consistent with slower-than-linear growth over that interval, while

$$\hat{\alpha}_i \approx 1$$

is consistent with approximately linear regret.

However, $\hat{\alpha}_i < 1$ at finite horizon does *not* prove asymptotic sublinearity, so all three diagnostics are interpreted jointly.

Oral-ready answer. We cannot prove an asymptotic property with finite simulation, so we estimated cumulative regret at increasing horizons. Our primary measure was $\hat{R}_T/T$, since that directly matches the definition of sublinear regret. We supplemented it with the incremental regret rate and a local log-log exponent to see whether the growth rate was actually decreasing.

Q3 — Which ϵ -greedy schedules show sublinear regret?

The ϵ -greedy policy is

$$A_t = \begin{cases} \arg \max_a Q_t(a), & \text{with probability } 1 - \epsilon_t, \\ \text{uniformly random arm,} & \text{with probability } \epsilon_t. \end{cases}$$

For the selected arm, the empirical mean is updated by

$$Q_{t+1}(A_t) = Q_t(A_t) + \frac{1}{N_{t+1}(A_t)} (R_t - Q_t(A_t)).$$

We applied the Q2 diagnostics to all six schedules.

Schedule	Diagnosis	Reason
$\epsilon = 0$	Not sublinear	No exploration; can lock permanently onto a wrong arm.
$\epsilon = 0.01$	Inconclusive	Finite-horizon behaviour does not establish either asymptotic sublinearity or a clear positive plateau.
$\epsilon = 0.10$	Not sublinear	Normalised regret and incremental rate plateau at a clearly positive level.
Polynomial decay	Sublinear	Normalised regret and late growth rate continue to fall; clearest sublinear evidence.
$\epsilon_t = 0.99^t$	Inconclusive	Mean performance is excellent, but run-level analysis reveals rare lock-in and near-linear failure.
$\epsilon_t = 0.90^t$	Not sublinear	Exploration vanishes very rapidly and the regret diagnostics plateau.

$\epsilon = 0$

There is no explicit exploration. An early noisy reward can make a suboptimal arm appear best. If the policy then remains greedy, it may never sample the true optimal arm enough to correct the mistake.

If it locks onto a suboptimal arm a ,

$$R_T \approx \Delta_a T,$$

so regret is linear.

$\epsilon = 0.01$ — **updated diagnosis: inconclusive**

In our finite-horizon experiment, the numerical result did not show a sufficiently clear positive asymptotic plateau by $T = 10^5$ to justify classifying the policy as non-sublinear from simulation alone.

Therefore, our numerical answer is

$\epsilon = 0.01$ is inconclusive from the finite-horizon experiment.

We should *not* claim that the simulation proves sublinearity. We should also not claim that our finite-horizon evidence clearly establishes linear regret.

The important distinction is:

- **Numerical conclusion for the workshop:** inconclusive.
- **Theoretical context:** a standard fixed nonzero ϵ continues random exploration forever, so asymptotically it is expected to incur linear regret.

These are different claims. Q3 asks us to investigate numerically, so our reported classification should reflect the finite-horizon numerical evidence.

$\epsilon = 0.10$

The continual exploration cost is much more visible. The normalised regret and incremental regret rate settle at a clear positive level, and the effective exponent is approximately one. Hence the numerical evidence is consistent with linear regret.

Polynomially decaying ϵ_t

The tested schedule is

$$\epsilon_t = \min \left\{ \left(\frac{k \log(t+1)}{t+1} \right)^{1/3}, 1 \right\}.$$

It satisfies the desired basic trade-off:

$$\epsilon_t \rightarrow 0,$$

so the fraction of random exploratory actions decreases, while exploration persists sufficiently long to correct poor early estimates.

Numerically,

$$\hat{\rho}_T \downarrow, \quad \hat{s}_i \downarrow, \quad \hat{\alpha}_i < 1.$$

This gives the clearest evidence of sublinear regret.

$\epsilon_t = 0.99^t$

The mean curve looks extremely good, but

$$\sum_{t=0}^{\infty} 0.99^t = \frac{1}{1 - 0.99} = 100 < \infty.$$

So only a finite expected amount of explicit exploration is scheduled.

Our run-level audit between $T = 30,000$ and $T = 100,000$ found:

- 499 of 500 runs with approximately zero late regret slope;
- 1 run with late regret slope approximately 0.84829.

Thus the mean hides a rare but serious lock-in failure. The finite-horizon result is therefore **inconclusive**.

$$\epsilon_t = 0.90^t$$

This schedule decays even faster:

$$\sum_{t=0}^{\infty} 0.90^t = \frac{1}{1 - 0.90} = 10.$$

Exploration disappears extremely quickly, so early mistakes are much more difficult to correct. The numerical regret diagnostics plateau, giving evidence of non-sublinear behaviour.

Oral-ready answer. Using the Q2 diagnostics, $\epsilon = 0$ and $\epsilon = 0.10$ are clearly not sublinear, while the polynomial schedule gives the clearest sublinear evidence. The updated result is that $\epsilon = 0.01$ is **inconclusive** from our finite-horizon study. Likewise, 0.99^t is inconclusive because its mean curve looks excellent but run-level analysis reveals rare lock-in. The 0.90^t schedule loses exploration too rapidly and behaves non-sublinearly in our experiment.

Q4 — Does UCB learn the true value of each arm over time?

UCB selects

$$A_t = \arg \max_a \left[Q_t(a) + c \sqrt{\frac{\log(t+1)}{N_t(a)}} \right].$$

The first term,

$$Q_t(a),$$

is the empirical value estimate.

The second term,

$$c \sqrt{\frac{\log(t+1)}{N_t(a)}},$$

is the exploration bonus. It is larger for arms that have been sampled less often.

To answer whether the true arm values are learned, we directly compared

$$Q_T(a)$$

with

$$q^*(a).$$

For each arm,

$$\text{RMSE}_T(a) = \sqrt{\frac{1}{M} \sum_{r=1}^M \left(Q_T^{(r)}(a) - q^*(a) \right)^2}.$$

We then averaged this over the ten arms.

At $T = 10^4$:

c	mean arm-wise RMSE
0	0.8981
1	0.7254
2	0.4090

Therefore,

$c = 2$ estimated the set of true arm values most accurately.

The reason is that larger c produces a stronger exploration bonus, so under-sampled arms are revisited more often and their empirical means are based on more observations.

This does not mean UCB samples all arms equally. Clearly inferior arms can still receive relatively few pulls because accurate estimation of every poor arm is not necessary for reward maximisation.

Oral-ready answer. We tested arm-value learning directly by comparing $Q_T(a)$ with the known $q^*(a)$ using per-arm RMSE over 300 reruns. The mean RMSE was 0.8981, 0.7254 and 0.4090 for $c = 0, 1, 2$, so $c = 2$ estimated all arm values best because its larger confidence bonus forced more exploration of under-sampled arms.

Q5 — Compare the regret of UCB and ϵ -greedy

We made the comparison fair by using:

- the same fixed q^* ,
- the same reward model,
- the same horizon,
- the same pseudo-regret definition,
- the same Monte Carlo averaging protocol.

At

$$T = 10^4,$$

the representative cumulative pseudo-regret values were

Policy	$\widehat{R}_T$
UCB $c = 1$	80.58
UCB $c = 2$	160.29
$\epsilon_t = 0.99^t$	221.98
$\epsilon = 0.01$	994.19

Thus,

UCB $c = 1$ achieved the lowest regret among the shown policies.

Why is UCB efficient?

ϵ -greedy explores uniformly. During an exploratory action, it may sample an arm that is already known to be poor.

UCB instead directs exploration according to uncertainty:

$$c \sqrt{\frac{\log(t+1)}{N_t(a)}}.$$

Thus under-sampled arms receive larger bonuses, while well-sampled arms receive smaller bonuses.

The most important comparison is

$$c = 1 \quad \text{vs.} \quad c = 2.$$

Q4 showed that $c = 2$ estimates all arm values better, but Q5 shows that $c = 1$ obtains lower regret.

Therefore,

best estimation $\neq$ best reward maximisation.

The extra exploration used by $c = 2$ improves knowledge of poor arms, but those additional pulls themselves incur regret.

Oral-ready answer. Using the same testbed and pseudo-regret definition, UCB $c = 1$ had the lowest regret, 80.58 at $T = 10^4$. UCB $c = 2$ had better arm-value estimates but higher regret, 160.29, because it paid for that extra information by exploring inferior arms more often. So the parameter that best identifies the whole bandit is not necessarily the one that maximises reward.

Q6 — Compare gradient bandit and UCB regret

Gradient bandit does not estimate $q^*(a)$ directly. Instead it learns preferences

$$H_t(a)$$

and converts them to action probabilities using softmax:

$$\pi_t(a) = \frac{e^{H_t(a) - \max_b H_t(b)}}{\sum_j e^{H_t(j) - \max_b H_t(b)}}.$$

The subtraction of $\max_b H_t(b)$ is only for numerical stability and does not change the probabilities.

The preference update is

$$H_{t+1}(a) = H_t(a) + \alpha(R_t - \bar{R}_t) (\mathbf{1}_{\{a=A_t\}} - \pi_t(a)).$$

At

$$T = 10^4,$$

our representative regret values were

Policy	$\hat{R}_T$
UCB $c = 1$	80.58
UCB $c = 2$	160.29
Gradient $\alpha = 0.1$	314.58
UCB $c = 0$	1408.07
Gradient $\alpha = 0.5$	1968.16

The best tested gradient-bandit case was

$$\alpha = 0.1,$$

with regret

$$314.58.$$

Relative to UCB $c = 1$,

$$\frac{314.58}{80.58} \approx 3.90.$$

Thus the best tested gradient-bandit configuration accumulated about

$$3.90 \times$$

the regret of UCB $c = 1$ on this experiment.

This is an empirical conclusion for this fixed stationary Gaussian testbed and finite horizon. It is not a universal statement that UCB always outperforms gradient bandit.

Oral-ready answer. We computed the same cumulative pseudo-regret for gradient bandit and UCB. The best tested gradient case, $\alpha = 0.1$, had regret 314.58, compared with 80.58 for UCB $c = 1$, so it was about 3.90 times larger. Our conclusion is only that UCB was more regret-efficient on this testbed and horizon.

Q7 — Gradient step-size sensitivity and convergence

To investigate whether gradient bandit reliably learns the optimal arm for arbitrary positive step size, we varied

$$\alpha$$

and monitored both

$$\pi_T(a^*)$$

and cumulative regret.

The results at $T = 10^4$ were

α	mean $\pi_T(a^*)$	$\hat{R}_T$
0.01	0.9952	1542.45
0.05	0.9991	357.39
0.10	0.9896	314.58
0.50	0.7899	1968.16
1.00	0.7200	2765.23
2.00	0.5500	5358.04
5.00	0.3233	11148.71

Therefore,

our numerical results do not support reliable convergence for arbitrary $\alpha > 0$.

Small α

A very small step size gives stable but slow learning.

For example,

$$\alpha = 0.01$$

ends with

$$\pi_T(a^*) \approx 0.9952,$$

but still accumulates

$$1542.45$$

regret.

The final policy is good, but the transient learning process is slow.

Moderate α

Moderate values, approximately

$$0.05\text{--}0.10,$$

give the best finite-horizon trade-off in our experiment.

Large α

The update contains

$$\alpha(R_t - \bar{R}_t).$$

When α is large, a single noisy reward can produce a large preference change.

Softmax then amplifies the preference difference into a highly concentrated policy:

early lucky reward $\rightarrow$ large preference $\rightarrow$ high action probability $\rightarrow$ less exploration $\rightarrow$ little chance to correct.

At

$$\alpha = 5,$$

the final behaviour was strongly bimodal:

- 32.33% of runs ended with

$$\pi_T(a^*) > 0.95;$$

- 67.67% ended with

$$\pi_T(a^*) < 0.05.$$

Thus the mean

$$0.3233$$

does not describe a typical run. Instead, some runs almost completely commit to the optimal arm while most commit almost completely to a wrong arm.

Oral-ready answer. Our experiments do not support reliable convergence for arbitrary positive constant α . Small α learns slowly, moderate α works best over this horizon, and large α makes the preference updates so aggressive that softmax can prematurely concentrate on an arm because of early noisy rewards. At $\alpha = 5$, about 32% of runs committed to the optimal arm and about 68% committed away from it.

Final Q1–Q7 oral summary

1. **Q1:** Sublinear regret means $\text{Regret}_T / T \rightarrow 0$, so average regret per step vanishes.
2. **Q2:** We use normalised regret, incremental regret rate and effective exponent across increasing horizons. Finite simulations provide evidence, not proof.
3. **Q3:** Polynomial decay gives the clearest sublinear evidence. $\epsilon = 0$ and $\epsilon = 0.10$ are not sublinear in our numerical study. $\epsilon = 0.01$ and 0.99^t are inconclusive. 0.90^t behaves non-sublinearly because exploration disappears too rapidly.
4. **Q4:** UCB $c = 2$ estimates the complete set of arm values best, with mean arm-wise RMSE 0.4090.
5. **Q5:** UCB $c = 1$ has the lowest regret, 80.58 at $T = 10^4$. Better identification of all arms is not the same as minimum regret.
6. **Q6:** The best tested gradient bandit, $\alpha = 0.1$, has regret 314.58, about 3.90 times that of UCB $c = 1$ on this experiment.
7. **Q7:** Gradient bandit is strongly step-size sensitive; large α can produce bimodal premature lock-in.